📘 SOC Notes – Day 7

🔐 Topic: Alerting & Incident Response Basics

🎯 Objective

Understand how alerts are generated and how SOC teams respond to security incidents.

🚨 1. What is an Alert?

An **alert** is a notification generated by a security system when suspicious or unusual activity is detected.

👉 In simple words:
An alert tells SOC analysts that *something might be wrong*.

⚠️ 2. Types of Alerts

- **Failed Login Alert**
  Triggered when multiple incorrect login attempts occur.
- **Malware Detection Alert**
  Raised when malicious software is detected in a system.
- **Unusual Network Activity Alert**
  Generated when abnormal traffic or behavior is observed.

🔄 3. Alert Workflow

1. Suspicious activity is detected
2. SIEM tool generates an alert
3. SOC analyst investigates the alert
4. Necessary action is taken

👉 This process helps in early detection of cyber threats.

🛡️ 4. What is Incident Response?

Incident Response (IR) is the process of identifying, managing, and resolving cybersecurity incidents.

👉 Goal:
Minimize damage and recover systems quickly.

🔁 5. Incident Response Steps

1. **Detection** – Identify the issue
2. **Analysis** – Understand the attack
3. **Containment** – Stop the spread
4. **Eradication** – Remove the threat
5. **Recovery** – Restore normal operations

💡 6. Example Scenario

- Multiple failed login attempts detected
- Alert is generated
- SOC analyst investigates
- Suspicious IP address is blocked